

odziv sechootjov (filtror) 1. reda na periodični vhod

$$z(t) = z_0 e^{i\omega t}, z_0 \in \mathbb{R}$$



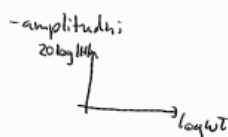
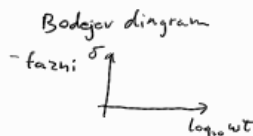
$$x(t) = x_0 e^{i\omega t}, x_0 \in \mathbb{R}$$

$$X(s) = H(s)Z(s), s = i\omega \rightarrow H(i\omega) = H(s=i\omega)$$

$$x_0 e^{i\omega t} \mathcal{L}(e^{i\omega t}) = H(i\omega) \mathcal{L}(e^{i\omega t}) z_0$$

$$(x_0 e^{i\omega t} z_0 H(i\omega)) \mathcal{L}(e^{i\omega t}) = 0$$

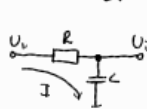
$$\frac{x_0}{z_0} e^{i\omega t} = H(i\omega)$$



$$\frac{x_0}{z_0} = |H(i\omega)|$$

$$\tan \delta = \frac{\operatorname{Im}(H(i\omega))}{\operatorname{Re}(H(i\omega))}$$

Pasivni filtri
Primer: LPF



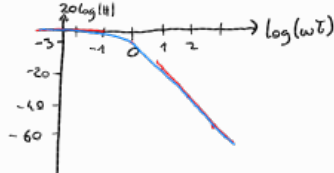
$$U_i = Z_c I = U_v \frac{1}{1+i\omega\tau}$$

$$I = \frac{U_i - 0}{R+Z_c}$$

$$\tau = RC \rightarrow H(i\omega) = \frac{1}{1+i\omega\tau}$$

$$|H|^2 = H H^* = \frac{1}{1+i\omega\tau} \frac{1}{1-i\omega\tau} = \frac{1}{1+(\omega\tau)^2}$$

$\omega\tau$	$ H ^2$	$10 \log H ^2$
$\gg 1$	$(\omega\tau)^{-2}$	$-20 \log \omega\tau$
$\ll 1$	1	0
1	$\frac{1}{2}$	$-10 \log 2 \approx -3 \text{ dB}$



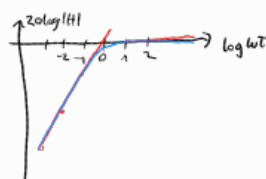
HPF

$$U_i = R I \rightarrow H(i\omega) = \frac{R}{R+Z_c} = \frac{1}{1+\frac{1}{i\omega\tau}} = \frac{i\omega\tau}{1+i\omega\tau}$$

$$I = \frac{U_i}{R+Z_c}$$

$$|H|^2 = \frac{(\omega\tau)^2}{1+(\omega\tau)^2}$$

$\omega\tau$	$ H ^2$	$10 \log H ^2$
$\gg 1$	1	0
$\ll 1$	$(\omega\tau)^2$	$20 \log(\omega\tau)$
1	$1/2$	-3 dB



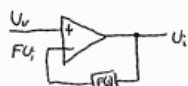
Aktivni filtri

• operacijski ojačevalnik

$$U_i = A(s)(U_+ - U_-)$$

	idealni	realni
$A(s)$	$A_0 \rightarrow \infty$	$A_0 H_{\text{diff}}, A_0 \sim 10^4 - 10^5$
Z_i, Z_o	$\rightarrow \infty$	$10^{10} - 10^{12} \Omega$
I_+, I_-	$\rightarrow 0$	$10^{-12} - 10^{-9} \text{ A}$
Z_i	$\rightarrow 0$	$\text{m}\Omega$

• negativna povratna zanka



Po dolgem času: $U_+ = U_-$

$$U_i = A(U_v - F U_i)$$

$$H = \frac{U_i}{U_v} = \frac{A}{1+AF} \approx \frac{1}{F}$$

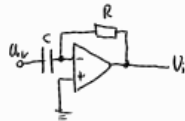
Primer:

diferenciator

$$U_i = k \frac{dU_v}{dt} / \omega$$

$$U_i(s) = k s U_v(s)$$

$$H(s) = k s = k i\omega$$



$$I_C = I_R \quad H = \frac{U_i}{U_v} = -\frac{R}{Z_C} = -i\omega R$$

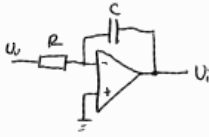
$$\frac{U_v}{Z_C} = -\frac{U_i}{R}$$

integrator

$$U_i = k \int U_v dt$$

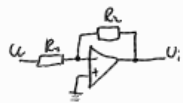
$$\dot{U}_i = k U_v$$

$$H(s) = \frac{k}{s} = \frac{k}{i\omega}$$



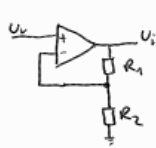
$$\frac{U_v}{R} = -\frac{U_i}{Z_C} \rightarrow H(i\omega) = -\frac{1}{i\omega R}$$

Invertirajoči ojačevalnik



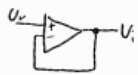
$$I_{R_1} = I_{R_2} \rightarrow \frac{U_v}{R_1} = -\frac{U_i}{R_2} \rightarrow H = -\frac{R_2}{R_1}$$

Neinvertirajoči ojačevalnik

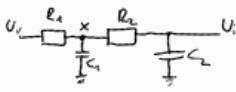


$$I_1 = I_2 \rightarrow H = 1 + \frac{R_2}{R_1}$$

Napetostni sledilnik



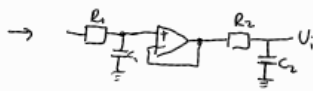
$$H=1$$



H_{LPF1} H_{LPF2}

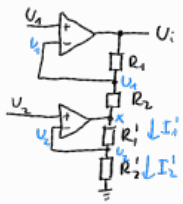
$$\left. \begin{aligned} x &= H_{LPF1} U_v \\ U_i &= H_{LPF2} x \end{aligned} \right\} U_i = H_{LPF1} H_{LPF2} U_v$$
$$H = H_{LPF1} H_{LPF2}$$

$$H_{LPF} \neq \frac{1}{1+i\omega\tau_1}, \text{ ker na izhod teče tok}$$



$$H_{LPF1} = \frac{1}{1+i\omega\tau_1} \checkmark$$

Odstevalnik



$$I'_1 = I'_2$$

$$\frac{x-U_2}{R'_2} = \frac{U_2}{R'_2}$$

$$x = U_2 \left(1 + \frac{R'_1}{R'_2}\right)$$

$$I_1 = I_2$$

$$\frac{U_i - U_1}{R_1} = \frac{U_1 - x}{R_2}$$

$$U_i = U_1 \left(1 + \frac{R_1}{R_2}\right) - x \frac{R_1}{R_2} =$$

$$= U_1 \underbrace{\left(1 + \frac{R_1}{R_2}\right)}_a - U_2 \underbrace{\left(1 + \frac{R'_1}{R'_2}\right)}_b$$

$$a=b$$

$$\rightarrow 1 + \frac{R_1}{R_2} = \frac{R_1}{R_2} \left(1 + \frac{R'_1}{R'_2}\right)$$

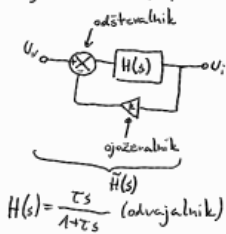
$$\boxed{\frac{R_1}{R_2} = \frac{R'_1}{R'_2}}$$

Pošlji pogoje za R_1, R_2, R'_1, R'_2 , da

se bo verjelo obnašalo kot

odstevalnik ($U_i \propto (U_1 - U_2)$)

Diagrami z neg. povr. zanko



a) $U_v \rightarrow H(s) \rightarrow U_i$

$U_v = k \cdot t$

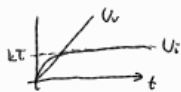
$U_v(s) = \frac{k}{s^2}$

$U_i = \frac{\tau s}{1 + \tau s} \cdot \frac{k}{s^2} = k\tau \left(\frac{A}{1 + \tau s} + \frac{B}{s} \right) = k\tau \left(\frac{1}{s} - \frac{\tau}{1 + \tau s} \right)$

$B=1$

$A=-\tau$

$U_i(t) = k\tau (1 - e^{-t/\tau})$



$U_i = H(U_v - kU_i)$

$U_i (1 + kH) = H U_v$

$\tilde{H} = \frac{H}{1 + kH} = \dots = \frac{\tau s}{1 + (1+k)\tau s}$

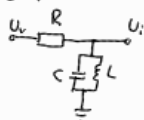
$\tilde{\tau} = (1+k)\tau \rightarrow \tilde{H} = \frac{1}{1+k} \frac{\tilde{\tau} s}{1 + \tilde{\tau} s}$
odvračalnik

$\tilde{U}_i(s) = \frac{1}{1+k} \frac{\tilde{\tau} s}{1 + \tilde{\tau} s} \cdot \frac{k}{s^2}$

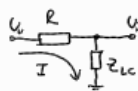
$\tilde{U}_i(t) = \frac{k\tilde{\tau}}{1+k} (1 - e^{-t/\tilde{\tau}}) = k\tau (1 - e^{-t/\tau})$

Filtri 2. reda (resonančna vezja)

BPF



ekvivalentno vezje



$U_i = Z_{LC} I = U_v \frac{Z_{LC}}{R + Z_{LC}}$
 $I = \frac{U_v}{R + Z_{LC}}$

$H = \frac{Z_{LC}}{R + Z_{LC}}$

$Z_{LC}^{-1} = Z_C^{-1} + Z_L^{-1} = i\omega C + \frac{1}{i\omega L} = sC + \frac{1}{sL} = \frac{s^2 LC + 1}{sL}$

$H = \frac{1}{1 + R/Z_{LC}} = \frac{1}{1 + R \frac{s^2 LC + 1}{sL}} = \frac{sL}{s^2 LC + sL + R} = \frac{s\omega_c}{s^2 + s\omega_c + \omega_0^2}$
 $\omega_c = \frac{1}{RC}$
 $\omega_0^2 = \frac{1}{LC}$

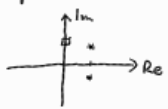
komentar:

$20 \log |H| = 20 \log |A| + 20 \log |P| - 20 \log |D|$

$H(s) = A \frac{(s-a_1)(s-a_2)\dots(s-a_m)}{(s-b_1)(s-b_2)\dots(s-b_n)}$
 $a_1, \dots$ ničle $H(s)$
 $b_1, \dots$ poli $H(s)$

graf polov in ničel

D... ničle
*...poli



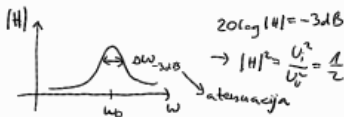
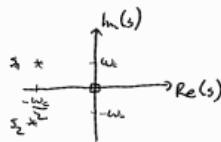
Imenovalca:

$s^2 + s\omega_c + \omega_0^2 = 0, \omega_0 > \omega_c$

$s_{1,2} = \frac{-\omega_c \pm \sqrt{\omega_c^2 - 4\omega_0^2}}{2}$

$= \frac{-\omega_c \pm i\sqrt{4\omega_0^2 - \omega_c^2}}{2}$

$\approx \frac{-\omega_c \pm i2\omega_0}{2} = -\frac{\omega_c}{2} \pm i\omega_0$



$H(s) = \frac{s\omega_c}{(s-s_1)(s-s_2)}$

$|H(\omega)| = \frac{|s\omega_c|}{|s-s_1||s-s_2|}, s = i\omega$

razdalja med s_1 in $(0, \omega)$

$|s-s_2| > |s-s_1|$

→ v okolici ω_0 se malo spreminja

obnašanje v okolici resonanče:

$\rightarrow \tilde{H} = \frac{1}{s-s_1}, |\tilde{H}| = \frac{1}{|i\omega-s_1|} = \frac{1}{|\frac{\omega_c}{2} + i(\omega-\omega_0)|} = \frac{1}{\sqrt{(\frac{\omega_c}{2})^2 + (\omega-\omega_0)^2}}$

$|\tilde{H}|^2 = \frac{1}{(\frac{\omega_c}{2})^2 + (\omega-\omega_0)^2} = \frac{1}{2} \frac{1}{(\frac{\omega_c}{2})^2}$

$\omega - \omega_0 = \pm \frac{\omega_c}{2} \rightarrow \omega = \omega_0 \pm \frac{\omega_c}{2}$

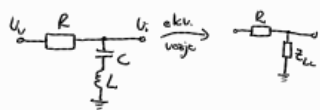
$\rightarrow \Delta\omega = \omega_2 - \omega_1 = \omega_c (= 2|Re(pole)|)$

Q...dobrota

$Q = \frac{\omega_0}{\Delta\omega_{3dB}} = \frac{\omega_0}{\omega_c} \gg 1$

$Q \gg 1 \dots$ ostra resonanca

Band-reject filter (notch filter)



$$H(s) = \frac{Z_{LC}}{R + Z_{LC}}$$

$$Z_{LC} = Z_L + Z_C = \frac{1}{sC} + sL = \frac{s^2 LC + 1}{sC}$$

$$H(s) = \frac{1 + s^2 LC}{s^2 LC + sRC + 1} = \frac{s^2 + \frac{1}{LC}}{s^2 + s\frac{R}{L} + \frac{1}{LC}} = \frac{s^2 + \omega_0^2}{s^2 + s\omega_c + \omega_0^2}$$

$$\omega_c^2 = \frac{1}{LC}$$

$$\omega_c = \frac{R}{L}$$

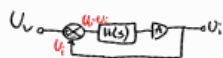
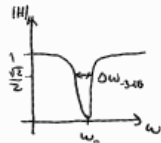
$$\Delta\omega_{3dB} = \omega_c = \frac{R}{L}$$

$$Q = \frac{L}{R} \frac{1}{\omega_c}$$

$$\text{Im: } s^2 + 2\xi\omega_0 s + \omega_0^2$$

$$\omega_c = 2\xi\omega_0$$

$$Q = \frac{1}{2\xi}$$



$$U_i = A H(s) (U_v - U_i)$$

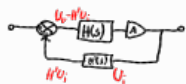
$$U_i (1 + A H(s)) = A H(s) U_v$$

$$H(s) = \frac{K}{\omega_c^2 + 2\xi\omega_0 s + 1}$$

$$\tilde{H} = \frac{U_i}{U_v} = \frac{A H}{1 + A H} = \frac{A K}{\frac{s^2}{\omega_c^2} + 2\xi\omega_0 s + 1 + A K} = \frac{A K}{1 + A K} \frac{1}{\frac{s^2}{\omega_c^2(1 + A K)} + 2\xi\omega_0 s + 1}$$

$$\tilde{H}(s) = ?$$

$$\Rightarrow \tilde{\xi} = \frac{\xi}{\sqrt{1 + A K}}, \quad \tilde{\omega}_0 = \omega_0 \sqrt{1 + A K}$$



$$H'(s) = 1 + T s$$

$$\tilde{H} = ?$$

$$U_i = A H (U_v - U_i)$$

$$\hat{H}(s) = \frac{A H}{1 + A H H'} = \frac{A K}{\frac{s^2}{\omega_c^2} + 2\xi\omega_0 s + 1 + A K + A K T s}$$

$$= \frac{A K}{1 + A K} \frac{1}{\frac{s^2}{\omega_c^2(1 + A K)} + \left(2\xi\omega_0 + \frac{A K}{1 + A K} T\right) s + 1}$$

$$\hat{\omega}_0 = \omega_0 \sqrt{1 + A K}$$

$$2 \frac{\xi}{\omega_c(1 + A K)} + \frac{A K}{1 + A K} T = 2 \frac{\hat{\xi}}{\hat{\omega}_0} \rightarrow T = 2 \frac{\hat{\xi}}{\omega_0 \sqrt{1 + A K}}, \quad \hat{\xi} = \frac{1}{\sqrt{2}} \text{ (optimalno dušenje)}$$

$$T = \frac{\sqrt{2}}{\omega_0 \sqrt{1 + A K}}$$